Year 11 Higher

REVISION PACK

Graphs 5

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 5 · Simultaneous equations (Lesson 3.1)

Q	Working	Answer
adding to remove a letter		
1.1	$4x + 3y = 18$ $2x - 3y = 0$ $\times 3:\ 12x + 9y = 54$ $\times 3:\ 6x - 9y = 0$ $\text{add}:\ 18x = 54$ $x = 3$ $4 \times 3 + 3y = 18$ $y = 2$	$x = 3, y = 2$
1.2	$5x + 2y = 18$ $3x - 2y = -2$ $\times 2:\ 10x + 4y = 36$ $\times 2:\ 6x - 4y = -4$ $\text{add}:\ 16x = 32$ $x = 2$ $5 \times 2 + 2y = 18$ $y = 4$	$x = 2, y = 4$
1.3	$2x + 5y = 13$ $4x - 5y = 11$ $\times 5:\ 10x + 25y = 65$ $\times 5:\ 20x - 25y = 55$ $\text{add}:\ 30x = 120$ $x = 4$ $2 \times 4 + 5y = 13$ $y = 1$	$x = 4, y = 1$

1.4	$3x + 2y = 18$ $x - 2y = -2$ $\times 2: \ 6x + 4y = 36$ $\times 2: \ 2x - 4y = -4$ $\text{add}: \ 8x = 32$ $x = 4$ $3 \times 4 + 2y = 18$ $y = 3$	$x = 4, y = 3$
1.5	$6x + y = 11$ $2x - y = -3$ $\text{add}: \ 8x = 8$ $x = 1$ $6 \times 1 + y = 11$ $y = 5$	$x = 1, y = 5$
1.6	$5x + 4y = 6$ $3x - 4y = 10$ $\times 4: \ 20x + 16y = 24$ $\times 4: \ 12x - 16y = 40$ $\text{add}: \ 32x = 64$ $x = 2$ $5 \times 2 + 4y = 6$ $y = -1$	$x = 2, y = -1$
multiplying first		
1.7	$5x + 3y = 12$ $2x - 4y = 10$ $\times 4: \ 20x + 12y = 48$ $\times 3: \ 6x - 12y = 30$ $\text{add}: \ 26x = 78$ $x = 3$ $5 \times 3 + 3y = 12$ $y = -1$	$x = 3, y = -1$

1.8	$3x + 4y = 10$ $5x - 3y = 7$ $\times 3:\ 9x + 12y = 30$ $\times 4:\ 20x - 12y = 28$ $\text{add}:\ 29x = 58$ $x = 2$ $3 \times 2 + 4y = 10$ $y = 1$	$x = 2, y = 1$
1.9	$2x + 5y = 13$ $3x + 2y = 3$ $\times 2:\ 4x + 10y = 26$ $\times 5:\ 15x + 10y = 15$ $\text{subtract}:\ 11x = -11$ $x = -1$ $2 \times -1 + 5y = 13$ $y = 3$	$x = -1, y = 3$
1.10	$4x + 5y = 13$ $3x - 2y = 4$ $\times 2:\ 8x + 10y = 26$ $\times 5:\ 15x - 10y = 20$ $\text{add}:\ 23x = 46$ $x = 2$ $4 \times 2 + 5y = 13$ $y = 1$	$x = 2, y = 1$
1.11	$6x + 5y = 16$ $4x - 3y = -2$ $\times 3:\ 18x + 15y = 48$ $\times 5:\ 20x - 15y = -10$ $\text{add}:\ 38x = 38$ $x = 1$ $6 \times 1 + 5y = 16$ $y = 2$	$x = 1, y = 2$

1.12	$3x + 2y = 11$ $4x - 5y = 7$ $\times 5:\ 15x + 10y = 55$ $\times 2:\ 8x - 10y = 14$ $\text{add}:\ 23x = 69$ $x = 3$ $3 \times 3 + 2y = 11$ $y = 1$	$x = 3, y = 1$
problems with two unknowns		
1.13	$2n + 3p = 2.25$ $n + 4p = 1.75$ $\times 4:\ 8n + 12p = 9$ $\times 3:\ 3n + 12p = 5.25$ $\text{subtract}:\ 5n = 3.75$ $n = 0.75$ $2 \times 0.75 + 3p = 2.25$ $p = 0.25$	**notebook: 0.750 KD** **pencil: 0.250 KD**
1.14	$5a + 2m = 3.3$ $3a + 4m = 3.8$ $\times 4:\ 20a + 8m = 13.2$ $\times 2:\ 6a + 8m = 7.6$ $\text{subtract}:\ 14a = 5.6$ $a = 0.4$ $5 \times 0.4 + 2m = 3.3$ $m = 0.65$	**apple: 0.400 KD** **mango: 0.650 KD**
1.15	$3c + c = 4.1$ $2c + 5c = 4.9$ $\times 5:\ 15c + 5c = 20.5$ $\text{subtract}:\ 13c = 15.6$ $c = 1.2$ $3 \times 1.2 + c = 4.1$ $c = 0.5$	**coffee: 1.200 KD** **cake: 0.500 KD**

1.16	$2s + j = 3.5$ $s + 3j = 3$ $\times 3:\ 6s + 3j = 10.5$ $\text{subtract}:\ 5s = 7.5$ $s = 1.5$ $2 \times 1.5 + j = 3.5$ $j = 0.5$	**sandwich: 1.500 KD** **juice: 0.500 KD**
1.17	$4p + 3e = 2.8$ $2p + 5e = 3.5$ $\times 5:\ 20p + 15e = 14$ $\times 3:\ 6p + 15e = 10.5$ $\text{subtract}:\ 14p = 3.5$ $p = 0.25$ $4 \times 0.25 + 3e = 2.8$ $e = 0.6$	**pencil: 0.250 KD** **eraser: 0.600 KD**
1.18	$3b + 2m = 10$ $b + 2m = 5$ $\times 2:\ 6b + 4m = 20$ $\times 2:\ 2b + 4m = 10$ $\text{subtract}:\ 4b = 10$ $b = 2.5$ $3 \times 2.5 + 2m = 10$ $m = 1.25$	**book: 2.500 KD** **magazine: 1.250 KD**
adding the equations		
1.19	$3x + y = 7$ $2x - y = 3$ $\text{add}:\ 5x = 10$ $x = 2$ $3 \times 2 + y = 7$ $y = 1$	$x = 2, y = 1$

1.20	$4x + y = 11$ $x - y = -1$ $\text{add}: \ 5x = 10$ $x = 2$ $4 \times 2 + y = 11$ $y = 3$	$x = 2, y = 3$
1.21	$2x + y = 8$ $5x - y = 13$ $\text{add}: \ 7x = 21$ $x = 3$ $2 \times 3 + y = 8$ $y = 2$	$x = 3, y = 2$
1.22	$x + y = 5$ $3x - y = 11$ $\text{add}: \ 4x = 16$ $x = 4$ $1 \times 4 + y = 5$ $y = 1$	$x = 4, y = 1$
1.23	$5x + y = 9$ $2x - y = -2$ $\text{add}: \ 7x = 7$ $x = 1$ $5 \times 1 + y = 9$ $y = 4$	$x = 1, y = 4$
1.24	$3x + y = 11$ $4x - y = 3$ $\text{add}: \ 7x = 14$ $x = 2$ $3 \times 2 + y = 11$ $y = 5$	$x = 2, y = 5$

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Q	Working	Answer
solving using graphs		
2.1	Lines cross at $(3, 3)$ Check: $2 \times 3 - 3 = 3$ and $3 + 3 = 6$	$x = 3, y = 3$
2.2	Lines cross at $(3, 5)$ Check: $3 + 2 = 5$ and $8 - 3 = 5$	$x = 3, y = 5$
2.3	Lines cross at $(2, 4)$ Check: $3 \times 2 - 2 = 4$ and $2 + 4 = 6$	$x = 2, y = 4$
2.4	Lines cross at $(2, 5)$ Check: $2 \times 2 + 1 = 5$ and $7 - 2 = 5$	$x = 2, y = 5$